Lagrangian Variable Cosmology final-v1.6: Correction of the Sound-Horizon Calibration and Re-evaluation of the v1.1 Modulation on the Corrected Pipeline

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Abstract

This paper is the formal successor to final-v1.5 in the LVC working-paper series. It records two inconsistencies in the reproduction pipeline used from v0.1 through v1.5, and the re-evaluation of the v1.1 modulation on the corrected pipeline. The inconsistencies are: (i) in the distance-prior integrals, the modulation factor $T(z)$ multiplied the dark-energy term of $H^2(z)$ only; (ii) the sound horizon entering the BAO likelihood was computed as $r_d = \theta_* D_{\rm M}(z_*)$, which is the sound horizon at the photon-decoupling epoch, $r_{\rm s}(z_*)$, and not the drag-epoch value $r_{\rm drag}$. On a data set of $N = 1596$ (DESI DR2 BAO, Pantheon+ without calibrators, Planck 2018 distance priors) the two inconsistencies change $\chi^2_{\Lambda\rm CDM}$ from 1405.56 (both corrected) to 1441.45 (both present), and change ΔBIC of the v1.1 locked profile with $\Omega_{\rm m}$ locked to 0.290653 from +14.38 to −38.77. On the corrected pipeline the $\Omega_{\rm m}$-locked v1.1 construction returns $\Delta\chi^2 = +21.76$ ($k = 2$) against ΛCDM. Releasing $\Omega_{\rm m}$ with the v1.1 profile and its remaining locks unchanged (denoted B0, $k = 3$) returns $\Delta\chi^2 = -10.69$, with $\Omega_{\rm m} = 0.3116$, $H_0 = 67.14$ and $r_{\rm drag} = 147.46$ Mpc. Across six variants of the distance-prior calibration and of the $r_{\rm drag}$ scale, $\Delta\chi^2$ of B0 ranges from −7.21 to −12.26 and that of the $\Omega_{\rm m}$-locked construction from +11.12 to +33.79. Sixteen neighbouring lock combinations, five profile shapes, parameter release up to six parameters, and the multi-rung ladder of v1.3 and v1.5 return no lower BIC than B0; the v1.5 three-rung construction returns $\Delta\chi^2 = -6.44$ against the ΔBIC = −41.83 reported there. The free-fit centre in $\theta = \ln(1+z)$ is $\theta_c = 0.4767$ ($\ln\varphi = 0.4812$). B0 returns $\Delta\chi^2 = -4.11$ with the DESI LRG2 bin removed. The CPL parametrisation ($k = 5$) returns $\Delta\chi^2 = -8.00$. DESI DR3 predictions of B0, the free-fit profile and CPL are tabulated. The values $\theta_c = \ln\varphi$, $w = (\ln\varphi)^3$ and $A = (\ln\varphi)^4$ were selected in v1.1 on the pipeline before correction; $\Delta\chi^2 = -10.69$ is an exploratory value.

1 Introduction

The working-paper series records a localised multiplicative modulation $T(z) = H(z)/H_{\Lambda\rm CDM}(z)$ of the late-time expansion rate. v1.1 [1] exhibited $T(z) - 1 = A\,\mathrm{sech}^2((\theta - \theta_c)/w)$ in $\theta = \ln(1+z)$ with $\theta_c = \ln\varphi$, $w = (\ln\varphi)^3$, $A = (\ln\varphi)^4$ and $\Omega_{\rm m} = 1/(1 + e^{3\ln\varphi/\varphi}) = 0.290653$. v1.3 [2] and v1.5 [3] examined a multi-rung generalisation with a non-minimal coupling. v1.5 recorded one inconsistency in the reproduction scripts and reduced the reported ΔBIC of the v1.1 construction from −30.31 to −11.82.

The present paper records a second inconsistency, located in the sound-horizon calibration of the BAO likelihood, and repeats the evaluation of the v1.1 family on a pipeline in which both are corrected. It does not introduce a new profile. Section 2 states the inconsistencies, the corrected pipeline, and the effect of each inconsistency on the v1.1 construction; it also records the re-evaluation of neighbouring candidates, of the multi-rung ladder and of the non-minimal coupling. Section 3 records the B0 configuration: its parameters, comparison with ΛCDM,

wCDM and CPL, BAO leave-one-out, the position profile in θ_c, and the sensitivity to a local H_0 prior. Section 4 records quantities computed from the B0 background without further fitting. Section 5 records a correction to the reconstruction of v1.1 Section 4. Section 6 records DESI DR3 predictions. Limitations are stated in Section 7.

The discipline of the paper is the same as in the preceding papers: phenomenology and reconstruction are kept separate, and the values reported are results of the stated procedures. The locks recorded are imposed inputs, not derivations from an action.

2 Corrected pipeline and validation

2.1 Two inconsistencies in the reproduction pipeline

(i) Modulation in the distance-prior integrals. In the scripts of v0.1–v1.4, the distance-prior integrals $D_{\rm M}(z_*)$ and $r_{\rm s}(z_*)$ used

$$H^2_{\rm old}(z) = H_0^2 \left[\Omega_{\rm m}(1+z)^3 + \Omega_r(1+z)^4 + (1-\Omega_{\rm m}-\Omega_r)\,T(z)^2\right], \tag{1}$$

whereas the BAO and supernova blocks used $H(z) = H_{\Lambda\rm CDM}(z)\,T(z)$. This was recorded in v1.5.

(ii) Sound horizon of the BAO likelihood. The sound horizon entering the BAO likelihood was computed as

$$r_d = \theta_*\, D_{\rm M}(z_*), \qquad \theta_* = 0.010409. \tag{2}$$

The quantity $\theta_* = r_{\rm s}(z_*)/D_{\rm M}(z_*)$ refers to the photon-decoupling epoch $z_* \simeq 1090$. The expression therefore returns $r_{\rm s}(z_*)$ ($\simeq 144.4$ Mpc at the Planck 2018 best-fit parameters), not the drag-epoch value $r_{\rm drag}$ (147.09 Mpc at the same parameters). The expression was used in all versions through v1.5, including $r_d = 144.52$ Mpc reported there. The values $r_d = 136.5$–145.3 Mpc reported in v8–v14, and the $N_{\rm eff}$ values inferred from them in v11, were obtained from the same expression.

2.2 Pipeline used in this paper

$T(z)$ multiplies $H(z)$ in all likelihood blocks. The quantities $r_{\rm drag}$, z_* and $r_{\rm s}(z_*)$ are computed with CAMB 2.0.4 [10] (neutrino mass sum 0.06 eV), tabulated on a grid in $(\Omega_{\rm b}h^2, \Omega_{\rm m}h^2) \in [0.0200, 0.0250] \times [0.10, 0.18]$ and interpolated with bicubic splines. At five test points the interpolated $r_{\rm drag}$, z_* and $r_{\rm s}(z_*)$ agree with direct CAMB evaluation to better than 2×10^{-6} (relative); changing the neutrino mass sum from 0.06 eV to 0 changes $r_{\rm drag}$ at the B0 best fit from 147.455 to 147.462 Mpc. Because $|T-1| < 10^{-15}$ for $z > 10$ (Table 2), these quantities are evaluated for the unmodulated background. The background in the distance integrals uses massless neutrinos with $N_{\rm eff} = 3.046$.

The distance priors $(R, \ell_A, \Omega_{\rm b}h^2)$ follow the definitions of Chen, Huang and Wang [7]. R and ℓ_A are multiplied by constants (0.999845 and 0.998286) fixed at the Planck 2018 fiducial cosmology ($H_0 = 67.36$, $\Omega_{\rm m} = 0.3153$, $\Omega_{\rm b}h^2 = 0.02237$) and applied identically to all models.

The data set is: DESI DR2 BAO [4] (13 measurements: one $D_{\rm V}/r_d$ at $z = 0.295$ and six $D_{\rm M}/r_d$, $D_{\rm H}/r_d$ pairs at $z = 0.510$, 0.706, 0.934, 1.321, 1.484, 2.330, with published correlations); Pantheon+ [5, 6] (1701 rows, of which 77 calibrator rows and 44 non-calibrator rows with $z_{\rm HD} \le 0.01$ are removed, leaving 1580; full STAT+SYS covariance; absolute magnitude profiled analytically); and the three distance priors. $N = 1596$. The SH0ES H_0, DESI DR1, BOSS DR12 and eBOSS DR16 are not included, in contrast to v1.1–v1.5; the numbers below are therefore not directly comparable to those papers.

ΔBIC is defined as $\Delta\chi^2 + (k - k_{\Lambda\rm CDM})\ln N$ with $k_{\Lambda\rm CDM} = 3$ $(H_0, \Omega_{\rm m}, \Omega_{\rm b}h^2)$ and $\ln N = 7.375$.

2.3 Reproduction and effect of each inconsistency

The original v1.5 script, run unchanged on the PP-excluded $N = 37$ data set of v1.1, returns $\chi^2 = 78.962$ (ΛCDM, $k = 3$) and 71.486 (v1.1 locks with $\Omega_{\rm m}$ locked, $k = 2$), as reported in v1.5. For ΛCDM the present pipeline returns $\chi^2 = 1405.56$ ($H_0 = 68.52$, $\Omega_{\rm m} = 0.2995$, $\Omega_{\rm b}h^2 = 0.02252$, $\chi^2_{\rm BAO} = 12.39$, $r_{\rm drag} = 147.44$ Mpc).

Table 1 records the v1.1 profile ($\theta_c = \ln\varphi$, $w = (\ln\varphi)^3$, $A = (\ln\varphi)^4$) with $\Omega_{\rm m}$ locked to 0.290653 ($k = 2$) and with $\Omega_{\rm m}$ free ($k = 3$), against ΛCDM, on the present data set, for the four combinations of (i) and (ii).

Configuration	$\chi^2(\Lambda\text{CDM})$	$\Delta\chi^2$ ($\Omega_{\rm m}$ locked)	ΔBIC	$\Delta\chi^2$ ($\Omega_{\rm m}$ free)
(i) and (ii) present	1441.45	-31.40	-38.77	-32.41
(ii) present, (i) corrected	1441.45	-19.40	-26.78	-19.72
(i) present, (ii) corrected	1405.56	$+22.90$	$+15.53$	-13.96
both corrected	1405.56	$+21.76$	$+14.38$	-10.69

Table 1: Effect of the two inconsistencies on the v1.1 profile, $N = 1596$. ΔBIC is given for the $\Omega_{\rm m}$-locked construction ($k = 2$).

2.4 Re-evaluation of the v1.1 family on the corrected pipeline

$\Omega_{\rm m}$ lock. With both inconsistencies corrected, the v1.1 construction with $\Omega_{\rm m}$ locked by Eq. (14) of v1.1 returns $\Delta\chi^2 = +21.76$ ($k = 2$). With $\Omega_{\rm m}$ released and all other v1.1 locks unchanged, it returns $\Delta\chi^2 = -10.69$ ($k = 3$; $\Omega_{\rm m} = 0.3116$, $H_0 = 67.14$, $\Omega_{\rm b}h^2 = 0.02254$, $r_{\rm drag} = 147.46$ Mpc, $\chi^2_{\rm BAO} = 5.2$). This configuration is denoted B0. The $\Omega_{\rm m}$ lock of v1.1 and the relation $\ln\varphi = \varphi\,\theta_{\rm eq}(\Omega_{\rm m})$ are not carried forward.

Neighbouring candidates. Sixteen lock combinations $A = (\ln\varphi)^a$, $w = (\ln\varphi)^b$ with $a, b \in \{2, 3, 4, 5\}$ and $\theta_c = \ln\varphi$ were evaluated with $\Omega_{\rm m}$ free ($k = 3$). The v1.1 combination $(a, b) = (4, 3)$ returns the lowest χ^2 ($\Delta\chi^2 = -10.69$); $(3, 4)$ and $(4, 4)$ return -10.10 and -9.71. At the v1.1 locks, replacing the profile by a Gaussian in θ, an exponential lift $\exp[A\,\mathrm{sech}^2]$, a Lorentzian and sech returns $\Delta\chi^2 = -11.1$, -10.6, -8.8 and -6.1; reversing the sign of A returns $+46.1$. The number of lock combinations evaluated was 16 ($2\ln 16 = 5.5$). Releasing parameters is recorded in Table 3.

Multi-rung ladder. The ladder of v1.3 and v1.5, $T - 1 = \sum_n A_n\,\mathrm{sech}^2((\theta - n\ln\varphi)/w)$ with $w = (\ln\varphi)^3$, was introduced as a self-similar lattice $\theta_n = n\ln\varphi$. On the corrected pipeline with $\Omega_{\rm m}$ free, the single rung returns $\Delta\chi^2 = -10.68$ ($k = 3$). Alternating-sign amplification $A_n = (-1)^{n+1}\varphi^{n-1}A_1$ with $n_{\rm max} = 2, 3, 4$ returns -5.74, $+5.49$, $+4.46$ ($k = 3$). Same-sign damping $A_n = \varphi^{-(n-1)}A_1$ with $n_{\rm max} = 2, 3$ returns -3.93, -0.21. The v1.5 three-rung lock $(A_2, A_3) = (-A_1, -\varphi A_1)$ returns -6.44, against ΔBIC $= -41.83$ reported in v1.5. With amplitudes free: A_2 free ($k = 4$) returns $A_2 = -0.0323$, $\Delta\chi^2 = -12.97$; A_2, A_3 free ($k = 5$) returns -14.49; A_2–A_4 free ($k = 6$) returns -14.54. The free A_2 has been reported as $+0.0295$ (v1.3), -0.0531 (v1.5) and -0.0323 (present). In the alternating family $A_8 = -1.557$ and $T = -0.56$ at the eighth rung. The ladder is not carried forward.

Non-minimal coupling. The coupling α_g of v1.3 and v1.5 was introduced to raise σ_8, which v1.3 reports as 0.787 for ΛCDM at the locked $\Omega_{\rm m} = 0.2907$. It is not carried forward.

Selection of the locks. The profile and locks of B0 were selected in v1.1 on the pipeline of Section 2.1. $\Delta\chi^2 = -10.69$ is therefore an exploratory value.

3 The B0 configuration

3.1 Definition and behaviour of $T(z)$

B0 is

$$H(z) = H_{\Lambda\mathrm{CDM}}(z; H_0, \Omega_\mathrm{m})\, T(z), \qquad T(z) = 1 + (\ln\varphi)^4 \operatorname{sech}^2\!\left(\frac{\theta - \ln\varphi}{(\ln\varphi)^3}\right), \qquad \theta = \ln(1+z), \tag{3}$$

with $\ln\varphi = 0.481212$, $(\ln\varphi)^3 = 0.111432$, $(\ln\varphi)^4 = 0.053622$. The free parameters are H_0, Ω_m, $\Omega_\mathrm{b}h^2$ ($k = 3$). Table 2 records $T - 1$.

z	$T-1$	z	$T-1$
0.30	4.1×10^{-3}	5	1.3×10^{-11}
0.62	5.4×10^{-2}	10	2.5×10^{-16}
1.0	4.6×10^{-3}	100	1.3×10^{-33}
2.0	3.3×10^{-6}	1090	3.6×10^{-52}

Table 2: $T(z) - 1$ for B0. $|T - 1| < 10^{-3}$ for $z > 1.18$, $< 10^{-6}$ for $z > 2.21$, $< 10^{-12}$ for $z > 5.92$ and below double-precision resolution for $z > 10.07$.

3.2 Comparison with ΛCDM, wCDM and CPL

Table 3 records fits on the corrected pipeline ($N = 1596$). wCDM has w_0 free; CPL has $w(a) = w_0 + w_a(1 - a)$ with w_0, w_a free; both are flat with $H_0, \Omega_\mathrm{m}, \Omega_\mathrm{b}h^2$ free.

Model	k	χ^2	$\Delta\chi^2$	ΔBIC	Parameters
ΛCDM	3	1405.56	0	0	$H_0 = 68.52$, $\Omega_\mathrm{m} = 0.2995$
v1.1, Ω_m locked	2	1427.33	+21.76	+14.38	$H_0 = 68.73$
B0	3	1394.88	−10.69	−10.69	$H_0 = 67.14$, $\Omega_\mathrm{m} = 0.3116$
B0, A free	4	1394.45	−11.11	−3.73	$A = 0.0446$
B0, A, w free	5	1394.16	−11.41	+3.34	$A = 0.0569$, $w = 0.0806$
A, θ_c, w free	6	1394.11	−11.46	+10.67	$A = 0.0593$, $\theta_c = 0.4767$, $w = 0.0777$
wCDM	4	1404.06	−1.51	+5.87	$w_0 = -0.9705$
CPL	5	1397.57	−8.00	+6.75	$w_0 = -0.849$, $w_a = -0.511$

Table 3: Fits on the corrected pipeline, $N = 1596$, $\theta_c = \ln\varphi$ and $w = (\ln\varphi)^3$ unless a parameter is listed as free.

3.3 BAO leave-one-out

Table 4 records $\Delta\chi^2$ of B0 against ΛCDM (both refitted, $k = 3$) with each DESI DR2 redshift bin removed in turn.

3.4 Position profile

Table 5 records $\Delta\chi^2$ against ΛCDM as a function of θ_c, with (A, w) locked to the B0 values (columns 2; $H_0, \Omega_\mathrm{m}, \Omega_\mathrm{b}h^2$ refitted) and with (A, w) free (columns 3–5; $H_0, \Omega_\mathrm{m}, \Omega_\mathrm{b}h^2$ refitted). The grid spacing is 0.025. The free-profile minimum is -11.45 at $\theta_c = 0.475$. The free-profile values at $\theta_c = 0.550$ and 0.575 have w at or near the lower bound $w = 0.02$.

Table 6 records the profile in A, with θ_c, w ($w \geq 0.03$), H_0, Ω_m, $\Omega_\mathrm{b}h^2$ refitted at each A. $\Delta\chi^2$ is against ΛCDM, which equals the value at $A = 0$ for any θ_c, w. The minimum is -11.46 at

Bin removed	$\Delta\chi^2$ (B0 − ΛCDM)
none	−10.69
BGS ($z = 0.295$)	−13.51
LRG1 ($z = 0.510$)	−8.00
LRG2 ($z = 0.706$)	−4.11
LRG3 ($z = 0.934$)	−9.50
QSO ($z = 1.321$)	−10.39
ELG2 ($z = 1.484$)	−10.77
Lyα ($z = 2.330$)	−10.79

Table 4: BAO leave-one-out for B0.

θ_c	$\Delta\chi^2$ (A, w locked)	$\Delta\chi^2$ (A, w free)	A	w
0.300	+7.18	−7.01	0.0307	0.351
0.350	−1.41	−7.56	0.0267	0.271
0.400	−6.08	−9.08	0.0311	0.166
0.425	−8.26	−10.00	0.0353	0.144
0.450	−10.13	−10.79	0.0412	0.120
0.475	−10.79	−11.45	0.0590	0.078
0.500	−9.72	−10.56	0.0477	0.093
0.525	−7.10	−8.69	0.0415	0.093
0.550	−3.14	−7.58	0.0787	0.020
0.575	+2.52	−6.37	0.1534	0.029
0.600	+10.08	−2.23	0.0131	0.252
0.650	+24.52	−0.90	0.0078	0.331
0.675	+26.60	−0.51	0.0058	0.363

Table 5: Profile in θ_c. $\ln\varphi = 0.4812$.

$A = 0.060$. Relative to the minimum, $\Delta\chi^2 < 4$ for the grid values $A = 0.02$ to 0.10; $\Delta\chi^2 = 6.22$ at $A = 0.01$ and 11.46 at $A = 0$.

3.5 Local H_0 prior

The likelihood of Section 2 contains no local H_0 measurement. The preferred values are $H_0 = 68.52$ (ΛCDM) and 67.14 (B0); the SH0ES value is 73.04±1.04 km/s/Mpc [9]. Adding a Gaussian term for the SH0ES value to the χ^2 gives $\chi^2 = 1423.11$ (ΛCDM; $H_0 = 68.84$, $\Omega_{\rm m} = 0.2955$; SH0ES term 16.28) and 1424.82 (B0; $H_0 = 67.55$, $\Omega_{\rm m} = 0.3063$; SH0ES term 27.83), so that $\Delta\chi^2$(B0 − ΛCDM) = +1.71. The statement of v8 that $H_0 = 73.04$ and θ_* are simultaneously satisfied was obtained with $r_d = \theta_* D_{\rm M}(z_*)$ (Section 2.1) and is not carried forward. The present modulation is not constructed to address the H_0 tension.

3.6 Sensitivity to pipeline choices

Table 7 records $\Delta\chi^2$ against ΛCDM (both refitted) for B0 and for the $\Omega_{\rm m}$-locked v1.1 construction when single choices of the pipeline are changed: the multiplicative calibration of ℓ_A (0.171% in the baseline) is removed or doubled; ℓ_A is removed from the distance priors (the covariance of $(R, \Omega_{\rm b}h^2)$ is the corresponding sub-block); $r_{\rm drag}$ is multiplied by 1.003 or 0.997.

A	$\Delta\chi^2$	θ_c	w
0.000	0	—	—
0.010	−5.24	0.4335	0.197
0.020	−8.49	0.4456	0.167
0.030	−10.25	0.4598	0.137
0.040	−11.09	0.4714	0.111
0.050	−11.40	0.4761	0.092
0.060	−11.46	0.4767	0.077
0.070	−11.43	0.4761	0.066
0.080	−11.38	0.4756	0.059
0.100	−11.24	0.4748	0.050

Table 6: Profile in A with θ_c, w free.

Variant	$\chi^2(\Lambda\mathrm{CDM})$	$\Omega_\mathrm{m}(\Lambda\mathrm{CDM})$	$\Delta\chi^2$ (B0)	$\Delta\chi^2$ (Ω_m locked)
baseline	1405.56	0.2995	−10.69	+21.76
no distance-prior calibration	1403.73	0.3013	−9.78	+28.24
ℓ_A calibration doubled	1407.59	0.2977	−11.60	+15.46
ℓ_A removed from distance priors	1400.93	0.3089	−7.21	+13.39
$r_\mathrm{drag} \times 1.003$	1403.00	0.3027	−9.10	+33.79
$r_\mathrm{drag} \times 0.997$	1409.18	0.2963	−12.26	+11.12

Table 7: Sensitivity of $\Delta\chi^2$ to single changes of the pipeline, $N = 1596$. B0 has $k = 3$; the Ω_m-locked construction has $k = 2$.

4 Quantities computed from the B0 background

No parameter is fitted in this section other than the amplitude $\sigma_8(0)$ in Section 4.2. Best-fit parameters are those of Table 3.

4.1 Effective dark energy

Defining $\rho_\mathrm{DE,eff} = E^2(z) - \Omega_\mathrm{m}(1+z)^3 - \Omega_r(1+z)^4$ with $E = H/H_0$ and $w_\mathrm{eff} = -1 - \frac{1}{3}\, d\ln\rho_\mathrm{DE,eff}/d\ln(1+z)$, Table 8 records the B0 and free-profile backgrounds. $\rho_\mathrm{DE,eff} > 0$ on $z \in [0, 3]$ for both. A CPL parametrisation matched to $E(z)$ of B0 on $z \in [0, 2.5]$ gives $(w_0, w_a) = (-0.80, -0.58)$ with rms $\delta E/E = 1.6\%$. For the CPL best fit of Table 3, $w(z) < -1$ for $z > 0.41$.

	B0	free profile
$w_\mathrm{eff}(z = 0)$	−1.001	−1.000
minimum w_eff (redshift)	−1.65 (0.51)	−1.98 (0.53)
maximum w_eff (redshift)	−0.42 (0.77)	−0.08 (0.71)
$w_\mathrm{eff} = -1$ crossing	$z = 0.64$	$z = 0.62$
$\rho_\mathrm{DE,eff}/\rho_\mathrm{DE,eff}(0)$ at $z = 0.62$	1.32	1.35
$\rho_\mathrm{DE,eff}/\rho_\mathrm{DE,eff}(0)$ at $z \geq 1.6$	1.00	1.00

Table 8: Effective dark energy of the modulated backgrounds. In both, $w_\mathrm{eff} > -1$ for z above the crossing and $w_\mathrm{eff} < -1$ below it.

4.2 Growth rate

The linear growth equation $D'' + (2 + d\ln H/dN)D' = \frac{3}{2}\Omega_{\rm m}(a)D$, $N = \ln a$, with $\Omega_{\rm m}(a) = \Omega_{\rm m}a^{-3}/E^2(a)$ and matter-era initial conditions $D(a_i) = a_i$, $D'(a_i) = a_i$ at $a_i = e^{-6}$, was integrated with $G_{\rm eff} = G$ (no coupling). $f\sigma_8$ was compared with the 16 measurements of Table 10 of v1.5 using diagonal errors; $\sigma_8(0)$ was profiled analytically. Table 9 records the result.

Background	profiled $\sigma_8(0)$	$\chi^2(f\sigma_8)$	$\chi^2(f\sigma_8)$ at $\sigma_8 = 0.811$
ΛCDM	0.824 ± 0.019	11.88	12.36
B0	0.837 ± 0.019	9.85	11.69
free profile	0.834	9.94	11.47
CPL	0.824	11.28	11.77

Table 9: $f\sigma_8$ fit with 16 points, diagonal errors, no coupling.

For B0, $S_8 = \sigma_8\sqrt{\Omega_{\rm m}/0.3} = 0.853$ at the profiled $\sigma_8(0)$; for ΛCDM, 0.823.

5 Correction to the reconstruction of v1.1 Section 4

The profile $v(\theta) \equiv T - 1 = A\,\mathrm{sech}^2((\theta - \theta_c)/w)$ is the only quantity of the reconstruction that enters the likelihood. With the field redefinition $\vartheta = 2\phi$ the potential of v1.1 becomes the standard sine–Gordon potential,

$$V(\phi) = m_\phi^2\,[1 - \cos\phi], \qquad m_\phi = m/2, \tag{4}$$

with the kink $\phi_{\rm kink}(\theta) = 4\arctan\exp[m_\phi(\theta - \theta_c)]$, for which $\sin(\phi_{\rm kink}/2) = \mathrm{sech}[m_\phi(\theta - \theta_c)]$. The identification

$$v(\theta) = \varepsilon^2\sin^2(\phi_{\rm kink}/2) = A\,\mathrm{sech}^2\left(\frac{\theta - \theta_c}{w}\right), \qquad A = \varepsilon^2, \quad w = 1/m_\phi = 2/m, \tag{5}$$

is algebraically equivalent to Eq. (9) of v1.1. Since $\sin^2(\phi/2) = V(\phi)/(2m_\phi^2)$, the identification can equivalently be written $v = (\varepsilon^2/2m_\phi^2)\,V(\phi_{\rm kink})$. The half-twist form of v1.1 is a reparametrisation of this potential; the period 4π of $V(\vartheta)$ does not enter the profile, the effective potential $V_{\rm eff}(v) = -(2/w^2)v^2(1 - v/A)$, the BPS relation $V_{\rm eff} = -\frac{1}{2}v_\theta^2$, or the relation $w = A/\ln\varphi$.

6 DESI DR3 predictions

6.1 Predicted offsets

Table 10 records the predicted D/r_d relative to ΛCDM (percent) at the DESI DR2 redshifts for B0, the free-profile fit and CPL, each at its best-fit parameters (Table 3), together with the DR2 uncertainty from the published covariance and the uncertainty obtained if the DR2 uncertainty is reduced by a factor 2 ($\sigma_{\rm DR3}$).

6.2 Expected BAO-only discrimination

Table 11 records the expected χ^2 obtained by taking the noiseless prediction of one model (the "truth") as data with DR3 uncertainties and the DR2 correlations, and refitting another model to it with H_0 and $\Omega_{\rm m}$ free (and w_0, w_a free for CPL; A, θ_c, w free for the free profile) at fixed $r_d = 147.45$ Mpc and $\Omega_{\rm b}h^2 = 0.0225$. The B0 column has the profile locked. The values scale as $(f/2)^2$ for a reduction factor f of the DR2 uncertainties (Table 12).

Observable	z	$\sigma_{\rm DR2}$	$\sigma_{\rm DR3}$	B0	free	CPL	(B0−CPL)/$\sigma_{\rm DR3}$
$D_{\rm V}/r_d$ (BGS)	0.295	0.94	0.47	+1.50	+1.24	−0.22	+3.6
$D_{\rm M}/r_d$ (LRG1)	0.510	1.24	0.62	+0.93	+0.84	−0.27	+1.9
$D_{\rm H}/r_d$ (LRG1)	0.510	1.95	0.98	−2.46	−2.22	−0.67	−1.8
$D_{\rm M}/r_d$ (LRG2)	0.706	1.04	0.52	−0.15	−0.30	−0.32	+0.3
$D_{\rm H}/r_d$ (LRG2)	0.706	1.71	0.85	−3.24	−2.79	−0.32	−3.4
$D_{\rm M}/r_d$ (LRG3)	0.934	0.71	0.35	−0.40	−0.37	−0.28	−0.3
$D_{\rm H}/r_d$ (LRG3)	0.934	1.09	0.55	−0.08	+0.32	+0.07	−0.3
$D_{\rm M}/r_d$ (QSO)	1.321	1.16	0.58	−0.24	−0.21	−0.16	−0.1
$D_{\rm H}/r_d$ (QSO)	1.321	1.53	0.77	+0.44	+0.34	+0.47	−0.0
$D_{\rm M}/r_d$ (ELG2)	1.484	2.48	1.24	−0.19	−0.17	−0.11	−0.1
$D_{\rm H}/r_d$ (ELG2)	1.484	4.00	2.00	+0.40	+0.29	+0.55	−0.1
$D_{\rm M}/r_d$ (Lyα)	2.330	1.36	0.68	−0.08	−0.08	+0.05	−0.2
$D_{\rm H}/r_d$ (Lyα)	2.330	1.17	0.59	+0.20	+0.13	+0.55	−0.6

Table 10: Predicted D/r_d relative to ΛCDM (percent); uncertainties in percent. The values of $\sigma_{\rm DR3}$ assume a factor-2 reduction relative to DR2.

Data generated by	B0 (locked)	free profile	ΛCDM	CPL
ΛCDM	35.79	0.00	0.00	0.00
CPL	26.35	0.00	0.87	0.00
B0	0.00	0.00	36.65	4.97

Table 11: Expected BAO-only χ^2 at $f = 2$.

6.3 Expected frozen-parameter discrimination

Because DESI DR3 contains DR2, a refit including DR3 contains the contribution of DR2 and the freedom of the refitted parameters. A comparison with all parameters fixed at the best fits of Table 3 (for CPL also w_0, w_a; r_d from CAMB at those parameters) is independent of both. Table 13 records the expected BAO χ^2 of a frozen model when the data equal the noiseless prediction of another model, with DR3 uncertainties obtained from DR2 by a factor f and the DR2 correlations. The values scale as $(f/2)^2$.

With $\Delta = \chi^2_{\rm frozen}({\rm B0}) - \chi^2_{\rm frozen}(\Lambda{\rm CDM})$ approximated as normal with mean $-\mu$ and variance 4μ when the data follow B0 (μ the entry of the first column), and mean $+\mu'$ and variance $4\mu'$ when the data follow ΛCDM (μ' the entry of the third column), the probabilities that the rule $\Delta > 0$ rejects B0 when B0 holds and retains B0 when ΛCDM holds are given in Table 14.

6.4 Conditions recorded before the DR3 release

The following conditions are recorded before the DR3 release. Parameters of B0 are those of Table 3.

(a) Frozen prediction. With the parameters of B0 and of ΛCDM fixed at the values of Table 3, B0 is not retained if its χ^2 on the DR3 BAO measurements exceeds that of ΛCDM.

(b1) Feature. With θ_c and w ($w \geq 0.03$) free, the feature is not retained if the profile likelihood in A on the DR3-included data set gives $\Delta\chi^2 < 4$ between $A = 0$ and the minimum. On the present data set this difference is 11.46 (Table 6).

(b2) Position. The φ lock is not retained if $\ln\varphi$ lies outside the interval of θ_c with $\Delta\chi^2 < 4$ in the DR3-included profile with (A, w) free. On the present data set the grid entries within $\Delta\chi^2 = 4$ of the minimum are $\theta_c \in [0.35, 0.525]$ (Table 5), which contains $\ln\varphi = 0.4812$.

f	ΛCDM refit to B0	CPL refit to B0	B0 refit to CPL	B0 refit to ΛCDM
1.0	9.16	1.24	6.59	8.95
1.3	15.48	2.10	11.13	15.12
1.5	20.62	2.80	14.82	20.13
2.0	36.65	4.97	26.35	35.79

Table 12: Expected BAO-only χ^2 as a function of the reduction factor f of the DR2 uncertainties.

f	B0 data, ΛCDM frozen	B0 data, CPL frozen	ΛCDM data, B0 frozen	CPL data, B0 frozen	CPL d
1.0	9.73	8.21	9.42	8.10	
1.3	16.44	13.87	15.92	13.69	
1.5	21.89	18.47	21.20	18.23	
2.0	38.92	32.84	37.68	32.41	

Table 13: Expected BAO χ^2 of a frozen model for data equal to the noiseless prediction of another model.

The following are reported irrespective of outcome: $\Delta\chi^2$ of B0 against ΛCDM and CPL on the DR3-included data set after refitting (k = 3, 3, 5); the frozen χ^2 of CPL; and the measured $D_{\rm H}/r_d$ at $z = 0.51$ and 0.706 against the entries of Table 10.

7 Limitations

Selection of the locks. The values $\theta_c = \ln\varphi$, $w = (\ln\varphi)^3$ and $A = (\ln\varphi)^4$ were selected in v1.1 on the pipeline before correction. The scan of Section 2.4 evaluated 16 lock combinations.

Parameter release. With A, θ_c and w free, $\Delta\chi^2 = -11.46$ at $\Delta\mathrm{BIC} = +10.67$ (Table 3). (A, w) lie on an extended ridge in the grid of Section 3.4: $(A, w) = (0.065, 0.07)$ and $(0.035, 0.11)$ with $\theta_c = \ln\varphi$ return $\Delta\chi^2 = -11.4$ and -10.6.

Dependence on single bins. With the LRG2 bin removed $\Delta\chi^2 = -4.11$ (Table 4).

Data set. Only DESI DR2 BAO is used; DR1, BOSS and eBOSS are not. No local H_0 prior is used in the main fits (Section 3.5). The $f\sigma_8$ comparison uses diagonal errors and a compilation of 16 measurements (Table 10 of v1.5) with overlapping samples.

Sound-horizon calibration. $r_{\rm drag}$, z_* and $r_{\rm s}(z_*)$ come from a CAMB 2.0.4 table with massive neutrinos (0.06 eV), while the distance integrals use massless neutrinos. The distance priors are calibrated by constants fixed at one fiducial cosmology.

CMB likelihood. The CMB enters through the compressed distance priors. No full Boltzmann likelihood is used. At the ΛCDM and B0 best fits of this data set, $\Omega_{\rm m}h^2 = 0.1406$ and 0.1405 respectively, against 0.1430 ± 0.0011 from Planck 2018 [8].

Sensitivity. $\Delta\chi^2$ of B0 ranges from -7.21 to -12.26 and that of the $\Omega_{\rm m}$-locked construction from $+11.12$ to $+33.79$ over the variants of Table 7. A change of 0.3% in $r_{\rm drag}$ changes the $\Omega_{\rm m}$-locked value by about 11 and the B0 value by about 1.6.

Lagrangian. No action is derived for $H(z)$.

f	μ	$P(\Delta > 0 \,\vert\, \mathrm{B0})$	μ'	$P(\Delta < 0 \,\vert\, \Lambda\mathrm{CDM})$
1.0	9.73	0.059	9.42	0.062
1.3	16.44	0.021	15.92	0.023
1.5	21.89	0.010	21.20	0.011
2.0	38.92	0.001	37.68	0.001

Table 14: Error probabilities of the frozen-parameter rule (normal approximation).

Forecast. Table 10 and Tables 11–12 assume a reduction factor of the DR2 uncertainties and use noiseless predictions at fixed r_d and $\Omega_{\mathrm{b}}h^2$.

8 Outlook

The external inputs recorded as next are DESI DR3 BAO (Section 6) and a full Boltzmann treatment of the CMB with the modulated background.

9 Conclusion

Two inconsistencies in the sound-horizon calibration and the distance-prior integrals are recorded; with both corrected, $\chi^2_{\Lambda\mathrm{CDM}} = 1405.56$ on $N = 1596$. The v1.1 profile with Ω_{m} locked returns $\Delta\chi^2 = +21.76$ ($k = 2$); with Ω_{m} free (B0) it returns $\Delta\chi^2 = -10.69$ ($k = 3$). The multi-rung ladder, the non-minimal coupling and the Ω_{m} lock are not carried forward. The free-fit centre is $\theta_c = 0.4767$. The profile and locks of B0 were selected before the correction. DESI DR3 predictions are tabulated in Section 6.

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A Numerical reproducibility

All numbers in Sections 2–6 were produced with a Python implementation of the pipeline of Section 2.2 (CAMB 2.0.4; Pantheon+ files `Pantheon+SH0ES.dat` and `Pantheon+SH0ES_STAT+SYS.cov` from the `PantheonPlusSH0ES/DataRelease` repository; DESI DR2 values as in the v1.5 script). Fits use differential evolution followed by Nelder–Mead refinement, with multiple starting points for the three-parameter fits. The original v1.5 script was run unchanged for the check of Section 2.3. The ΛCDM result ($\chi^2 = 1405.56$, $H_0 = 68.52$, $\Omega_{\mathrm{m}} = 0.2995$, $\chi^2_{\mathrm{BAO}} = 12.39$) was compared with an independent earlier implementation of the same pipeline (1405.4, 68.4, 0.301, 12.5).

Code availability. The reproduction script for this paper, `lvc_f-v1_6_reproduce.py`, is provided as a supplementary file under the MIT licence. It requires Python and the dependencies listed in the script, including CAMB 2.0.4 for the reported numerical results. The original v1.5 script is retained unchanged for the check of Section 2.3.

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